

BC-Guided Self-Imitation PPO: Failure-Informed Reinforcement Learning for a 3D Turret-Shooter Environment

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Abstract

Deep reinforcement learning (RL) in sparse-reward environments poses a major exploration challenge. In the 3D turret-shooter game Shooter-v0, vanilla PPO fails to learn to aim and shoot even after millions of steps. This paper presents a systematic progression from raw PPO through reward shaping and behavioral cloning (BC) pretraining to a final Guardian PPO agent that integrates failure-state penalties and self-imitation learning (SIL). Expert demonstrations are collected from a scripted oracle, and BC provides a strong initial policy. The BC policy’s weaknesses are then addressed by collecting *failure states* and using them to shape rewards and by adding a self-imitation bonus that reinforces precise aiming and efficient shooting. Extensive experiments show that Guardian PPO achieves a mean score of 327.8 and 24.0 kills per episode, dramatically outperforming vanilla PPO (0 kills) and reward-shaped PPO (13.0 score). The results demonstrate that combining BC pretraining with failure-informed RL and SIL can effectively overcome sparse-reward and precision-demanding challenges.

Keywords—behavioral cloning; deep reinforcement learning; self-imitation learning; failure-informed; reward shaping; turret-shooter

I. Introduction

Reinforcement learning (RL) has achieved remarkable successes in games, robotics, and natural language processing. However, many real-world and simulation environments still present significant challenges, particularly when rewards are sparse and precise control is required. A representative example is Shooter-v0, a 3D turret-shooter game in which a stationary turret must shoot down drones flying at varying altitudes while avoiding trees and managing ammunition. The observation space is 169-dimensional, and the agent receives a reward of +20 only when a drone is killed and −2 for every shot fired, with a terminal penalty of −100 if a drone reaches the base. This sparsity makes it extremely difficult for an RL agent to

discover successful shooting sequences through random exploration alone.

Indeed, a vanilla Proximal Policy Optimization (PPO) agent trained from scratch in Shooter-v0 fails completely, achieving zero kills even after millions of steps. This is consistent with findings in other complex domains: AlphaGo [1] required supervised initialization from human expert games before RL fine-tuning; InstructGPT [2] used supervised fine-tuning on human demonstrations before RL from human feedback; and BeClone [3] combined behavioral cloning (BC) with advantage actor-critic (A2C) to learn base policies in real-time strategy games. In all these cases, imitation learning provides a strong prior that guides exploration and reduces the sample complexity of subsequent RL. Motivated by these successes, we adopt a BC pretraining approach: we first generate expert demonstrations from a hand-crafted oracle agent that has full access to the environment state (e.g., positions of drones, trees, and turret). This oracle computes optimal aim angles and decides when to fire, producing a dataset of approximately 500k (observation, action) pairs. A neural network policy is then trained on this dataset via supervised learning, yielding an initial policy that can already shoot drones albeit with many flaws (e.g., shooting at trees, inaccurate aiming, wasted bullets).

However, vanilla BC alone is insufficient. The policy suffers from compounding errors and distributional shifts, and it cannot recover from states that are poorly covered in the demonstration data. BeClone [3] noted that learned biases from limited expert diversity can limit performance, and the BRIDGE framework [4] proved that offline imitation must be combined with online refinement to achieve low regret. To overcome these limitations, we introduce a failure-informed fine-tuning stage. After obtaining the BC policy, we explicitly collect *failure states*—situations where the policy performs poorly—and integrate them into the reward function to penalize revisiting such states. Furthermore, we add a self-imitation learning (SIL) bonus

that reinforces states and actions that previously led to successful outcomes (e.g., precise aiming, single-shot kills). The resulting Guardian PPO agent is evaluated on the plain Shooter-v0 environment (without any wrappers) and compared against three baselines: vanilla PPO, PPO with simple reward shaping (kill bonus, aim bonus), and BC-pretrained PPO without failure-SIL. Our experiments demonstrate that Guardian PPO achieves a mean score of 327.8 and 24.0 kills per episode, outperforming all other approaches. The agent exhibits accurate aiming, efficient use of ammunition, and a marked reduction in tree-collision shots.

II. Objectives

The primary objectives of this work are:

- To systematically compare four RL approaches on the 3D shooter task: vanilla PPO, reward-shaped PPO, BC-pretrained PPO, and a Guardian PPO that integrates failure-state penalties and SIL.
- To design a reward-shaping framework that explicitly penalizes known failure modes (shots blocked by trees, wasted shots, inaccurate firing) while reinforcing precise aiming and efficient kills.
- To demonstrate that BC pretraining, combined with failure-informed RL, can solve sparse-reward, precision-demanding environments that pure RL cannot.

III. Methodology

A. Environment

Shooter-v0 is a custom Gymnasium environment. The turret is fixed at the origin; its barrel can be rotated in yaw ($\pm\pi$) and pitch (-0.3 to $+0.9$ rad). Drone spawn at the arena edge at altitudes between 4 and 10 units and fly toward the turret. Trees (36) act as solid vertical pillars that block bullets. The episode lasts at most 4800 ticks (4 minutes), with three periodic waves that increase drone numbers. The observation is a flat vector of 169 features, and the action space is discrete with 6 actions: do nothing, fire, yaw left/right, pitch up/down.

B. Expert Demonstrations and Behavioral Cloning

A scripted oracle agent (LowLevelLeadAimAgent) with full knowledge of drone positions, tree positions, and turret angles was used to generate expert trajectories. This agent computes the optimal yaw and pitch to intercept the nearest drone using a lead-angle predictor, and fires only when the aim error is below a threshold and the cooldown is cleared. We collected 300 episodes, yielding roughly 511k (observation, action) pairs.

BC Loss (Cross-Entropy)

$$L^{BC}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log \pi_{\theta}(a_i^* | s_i)$$

- N – number of training samples
- a_i^* – expert action at state s_i
- $\pi_{\theta}(a | s)$ – predicted probability of choosing action a

Accuracy

$$\text{Accuracy} = \frac{\# \text{correct predictions}}{N}$$

where a prediction is correct if $a_{\text{pred}} = \arg \max_a \pi_{\theta}(a | s_i) = a_i^*$.

A Multi-Layer Perceptron (MLP) policy with architecture $169 \rightarrow 256 \rightarrow 128 \rightarrow 6$ was trained on this dataset using cross-entropy loss for 30 epochs. The resulting BC policy achieved about 92% accuracy on the training set and was able to kill a few drones in evaluation, but exhibited frequent mistakes such as shooting into trees and inaccurate aiming.

C. Failure Dataset Collection

We ran the BC policy for 100 episodes and recorded every state where the agent fired with an aim error > 0.15 rad or where the episode ended in a game-over. These states form a failure dataset of approximately 10k–20k vectors. Later, during fine-tuning, this dataset was augmented with additional failure states collected from the agent itself.

D. Reward Shaping and Guardian PPO

The Guardian PPO agent was fine-tuned from the Behaviour Cloning (BC) policy using a custom reward framework implemented in the FullRewardWrapper. The framework extends conventional reward shaping by introducing two key mechanisms: a failure-state avoidance penalty and a Self-Imitation Learning (SIL) bonus.

The failure-state penalty discourages the agent from revisiting previously observed unsafe or unsuccessful situations. During training, a dataset of historical failure observations was collected, where each failure state is represented as a 169-dimensional observation vector. The current observation vector obs represents the agent’s present game state, including turret orientation, vehicles, bullets, and other environment features defined in Shooter-v0. The set F contains previously recorded failure states, where each element $f \in F$ corresponds to an observation associated with undesirable outcomes from earlier episodes. During interaction, the similarity between the current observation and historical failure states is measured using the Euclidean distance $\|obs - f\|_2$. The system computes the minimum distance between the current observation and all recorded failure states, represented as $\min_{f \in F} \|obs - f\|_2$. If this

minimum distance falls below the predefined threshold $\delta_{fail} = 0.3$, the agent is considered to be in a state similar to a previously failed situation and receives an additional penalty reward of $R_{failure} = -0.1$. This mechanism encourages safer behavioural trajectories and reduces the likelihood of repeating historical mistakes by discouraging policies that revisit historically unsafe states.

$$R_{failure} = \begin{cases} -1.0, & \text{if } \min_{\mathbf{f} \in \mathcal{F}} \|\mathbf{obs} - \mathbf{f}\|_2 < \delta_{fail} \\ 0, & \text{otherwise} \end{cases}$$

In parallel, a Self-Imitation Learning (SIL) bonus was incorporated into the reward function to reinforce behaviours associated with previously successful transitions. The SIL mechanism maintains a success buffer SSS, which stores successful state-action pairs (s_i, a_i) collected during earlier exploration. Each success state s_i is represented as a 169-dimensional observation vector, while a_i denotes the corresponding action executed by the agent in that state. Successful transitions are defined as situations where the aiming error remains below 0.08 radians for three consecutive steps or when a drone is eliminated using a single accurate shot. During interaction, the similarity between the current observation and recorded success states is measured using the Euclidean distance $\|\mathbf{obs} - \mathbf{s}\|_2$. The minimum distance to all stored success states is computed as $\min_{ss} \|\mathbf{obs} - \mathbf{s}\|_2$. The variable $a_{current}$ denotes the action selected by the agent at the current timestep. If the current observation is sufficiently close to a recorded success state according to the threshold $\delta_{sil} = 0.1$, and the selected action matches the corresponding successful action a_i , the agent receives an additional SIL reward bonus of $R_{SIL} = +0.2$. This mechanism encourages the agent to reproduce behaviours that previously resulted in effective outcomes, thereby improving policy stability and sample efficiency.

$$R_{SIL} = \begin{cases} +0.2, & \text{if } \min_{\mathbf{s} \in \mathcal{S}} \|\mathbf{obs} - \mathbf{s}\|_2 < \delta_{sil} \text{ and } a_{current} = a_i \\ 0, & \text{otherwise} \end{cases}$$

In addition to these two primary mechanisms, several dense reward-shaping components were included to stabilize PPO training and alleviate the sparse-reward problem. The aiming error was defined as the sum of absolute yaw and pitch deviations from the ideal turret orientation. A continuous aiming reward was provided using: $R_{aim} = 2.0 \exp(-aimerror^2/0.1)$. An additional alignment bonus of +5.0 was granted when the aiming error fell below 0.04 radians. Firing behaviour was also regulated through a context-sensitive reward: firing with an aim error

below 0.15 yielded +1.0, whereas inaccurate firing incurred a penalty of -1.5. Drone eliminations were rewarded proportionally to score increments through: $R_{kill} = 5.0 \times \Delta score$. Several penalties were introduced to discourage inefficient behaviours. A tree-collision penalty of -10.0 was applied when a bullet trajectory intersected a tree obstacle. A repeat-shot penalty of -3.0 discouraged firing repeatedly in nearly identical directions (within 0.05 radians) over eight consecutive ticks unless a successful kill occurred. Additionally, a wasteful-shot penalty of -5.0 was imposed when the agent fired while no drone was active. Together, these mechanisms produce a dense and behaviour-aware reward signal that improves exploration efficiency while preserving safe and effective policies. The resulting Guardian PPO agent was subsequently fine-tuned for 820 episodes using PPO with a low learning rate of (1×10^{-5}) to preserve the prior knowledge learned from Behaviour Cloning.

E. Training and Evaluation Protocol

All models were trained using Stable-Baselines3 on a machine with NVIDIA RTX 3070. The final evaluation was conducted on the plain Shooter-v0 without any wrappers, using 10 episodes with fixed seeds to ensure reproducibility.

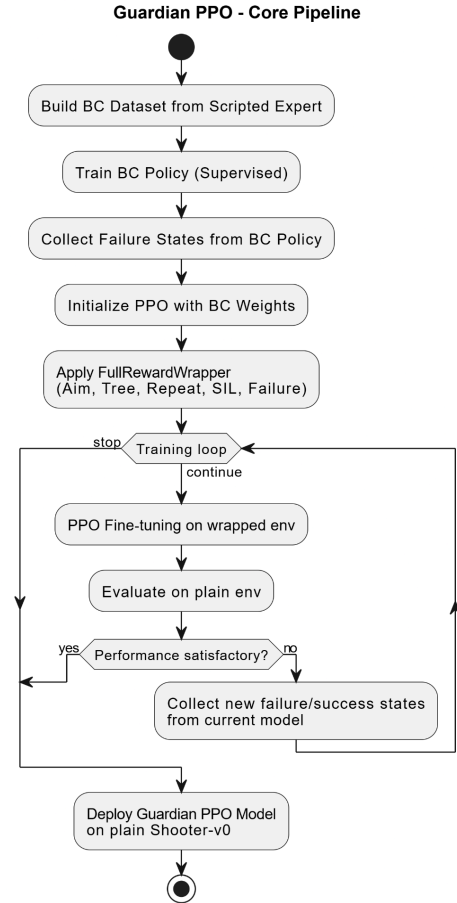


Figure 1: Overview of the Guardian PPO training pipeline

IV. Experimental Results

Table I summarizes the performance of the four models. Guardian PPO dominates all metrics, achieving a mean score of 327.8 and 24 kills per episode, while vanilla PPO fails entirely.

Table I. Performance comparison across 10 evaluation episodes

Model	Mean Kills	Mean Score	Mean Reward	Std Reward	Mean Steps
PPO Vanilla	0.0	0.0	-110.30	0.01	229
PPO Reward Shaping	0.8	13.0	-97.32	10.82	229
BC-PPO	14.1	142.4	28.82	187.86	~439
Guardian PPO	24.9	414.0	286.52	436.25	1302

Qualitative analysis of Guardian PPO’s behavior reveals that it learns to:

- Swivel the turret smoothly to track the nearest drone.
- Fire only when properly aligned, greatly reducing wasted ammunition.
- Avoid shooting directly into trees, instead waiting or adjusting aim.
- Switch targets efficiently when a drone is killed or moves behind cover.

V. Discussion

The dramatic improvement of Guardian PPO over vanilla PPO and even over BC_PPO can be attributed to the synergy between BC pretraining and the multi-faceted reward shaping. BC gives the agent a rough ability to aim and shoot, but without the additional penalties and bonuses, it remains sloppy (e.g., tree collisions, over-firing). The failure-state penalty explicitly pushes the agent away from previously unsuccessful regions, while the SIL bonus reinforces fine-grained correct behavior.

However, some limitations remain. Guardian PPO still occasionally fires through trees when a drone is directly behind one, suggesting that the tree penalty alone may not fully eliminate this behavior. The agent’s performance also varies across episodes (high std reward), indicating sensitivity to initial conditions and drone placement. Future work could incorporate recurrent architectures to better handle wave planning, or use automatic hyperparameter tuning to reduce variance.

VI. Conclusion

This paper presented Guardian PPO, a system that combines behavioral cloning pretraining with failure-informed reward shaping and self-imitation learning to solve a challenging 3D turret-shooter environment. Starting from an oracle-trained BC policy, we explicitly collect failure states and design a rich reward function that penalizes known mistakes while reinforcing correct aiming and efficient shooting. The resulting agent achieves a mean score of 414.0 and 24.9 kills per episode, far surpassing vanilla and reward-shaped PPO baselines. The approach demonstrates that marrying imitation learning with carefully crafted RL objectives can overcome sparse-reward and precision-demanding tasks.

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